UNIT 6 · STANDARD 6.EE.3

Simplify Algebraic Expressions

Lesson 6-7

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$5x$ and $3x$ are like terms (both x); $5x$ and $5y$ are NOT (different variables); $2x$ and $2x^2$ are NOT (different powers)

Like Terms

Terms with the same letter, like $2x$ and $5x$.

$8m + 3m = 11m$ — add the coefficients ($8 + 3 = 11$) and keep the variable (m)

Combine

To add or subtract terms with the same letter into one.

$2a + 5a + 3$ simplifies to $7a + 3$ — the two a-terms combine, the constant stays

Simplify

To write an expression in its shortest form.

In $9x$, the coefficient is 9 — it tells you 'nine groups of x'; if $x = 2$, then $9x = 18$

Coefficient

The number in front of a letter, like the 3 in $3x$.

In $3x + 5 - 2x$, the terms are $3x$, 5, and $2x$.

Term

A single number, variable, or product of them, separated by + or – signs.

$3(x + 4) = 3\cdot x + 3\cdot 4 = 3x + 12$.

Distributive property

Multiplying a sum by a number means multiplying each part by that number.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can simplify algebraic expressions by combining like terms.

1 Notice

The music studio is placing supply orders for three rehearsal rooms.

2 Key idea

I can simplify algebraic expressions by combining like terms.

3 Words I'll use

Like Terms **Combine** **Simplify** **Coefficient** **Term** **Distributive property**

🤔 Think About It

- Why can you add the microphones together but not mix microphones with stands?
- What makes microphones and stands 'different types' of items?
- How is this like combining like terms in algebra?

👀 Watch

Watch your teacher model one example. Jot what you see.

🤝 We try

Solve the next one together as a class.

✏️ You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Three rooms need microphones and stands. Why can you add all the microphones together (4 + 2 + 6) but not mix microphones with stands?

Level 1 support

Start here: You can only combine items of the same type, just like like terms.

💡 **Need a hint?**

Sentence starters (optional)

I can add the microphones because they are the ____ type.

Puedo sumar los micrófonos porque son del ____ tipo.

I cannot mix microphones and stands because they ____.

No puedo mezclar micrófonos y soportes porque ____.

Word bank: Like Terms | Combine | Simplify | Coefficient

Listen for: *Students explain microphones combine with microphones (4 + 2 + 6 = 12) but not with stands, connecting this to combining like terms in algebra.*

Level 2

How is mixing microphones and stands like trying to add 3x and 5? Generalize the rule for which terms can combine.

- Mixing microphones and stands is like adding 3x and 5 because ____.
- In general, terms combine only when they have the ____.

EXPLORE

To simplify 6x + 4 + 2x + 7, which are the like terms, and how do you combine them?

Level 1 support

Start here: Combine the x-terms by adding coefficients; combine the constants separately.

💡 **Need a hint?**

Sentence starters (optional)

The like terms are ___ and ___.

Los términos semejantes son ___ y ___.

I combine them by ___ to get ___.

Los combino al ___ para obtener ___.

Word bank: Like Terms | Combine | Simplify | Coefficient

Listen for: *A strong answer combines 6x + 2x = 8x and 4 + 7 = 11 to get 8x + 11, adding coefficients of like terms and keeping the variable.*

Level 2

A classmate simplifies 5x + 3 + 2y + 4x as 11xy + 3. Critique the error and give the correct simplified form.

- The error is ___ because 5x and 2y are not ___.
- The correct simplified form is ___ because I combine only ___.

CONNECT

The festival order is (8m + 5) + (3m + 7) + (4m + 2), where m is microphones per stage. What does the simplified expression tell you?

Level 1 support

Start here: Combine the m-terms and the constants to read the total at a glance.

💡 **Need a hint?**

Sentence starters (optional)

The total simplifies to ____m + ____.

El total se simplifica a ____m + ____.

The ____m tells us ____ and the constant tells us ____.

El ____m nos dice ____ y la constante nos dice ____.

Word bank: Like Terms | Combine | Simplify | Coefficient

Listen for: *Students simplify to 15m + 14, explaining 15m is the total cables tied to microphones and 14 is the extra fixed cables across stages.*

Level 2

Why is 15m + 14 easier to use for ordering than the original three grouped expressions? Justify how simplifying makes the real problem clearer.

- The simplified form is easier because ____.
- If each stage had 6 microphones, the total cables would be ____, which is faster to find because ____.

Guided Examples

Example 1

Simplify $7x + 3x$.

Solution: $7x + 3x = 10x$. Add the coefficients: $7 + 3 = 10$.

Answer: A. $10x$

Example 2

Simplify $9n + 4 + 2n$.

Solution: $9n + 2n = 11n$. The constant 4 has no like term, so it stays. Result: $11n + 4$.

Answer: A. $11n + 4$

Example 3

Which pair are like terms?

Solution: $6x$ and $2x$ are like terms because they both have the same variable (x) raised to the same power (1).

Answer: A. $6x$ and $2x$

Write About the Math

The Writing Revolution

I can explain my steps using the words like terms, combine, simplify, and coefficient.

1. Kernel Sentence

subject + verb

Model: Simplify is to write an expression in its shortest form.
Simplificar es escribir una expresión en su forma más corta.

Write a kernel sentence about simplify. Use a subject and a verb.
Escribe una oración base sobre simplificar. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Simplify matters in math
Simplificar importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*

Simplify matters in math because ____.
Simplificar importa en matemáticas porque ____.

____

but *pero*

Simplify matters in math, but ____.
Simplificar importa en matemáticas, pero ____.

____

so *entonces*

Simplify matters in math, so ____.
Simplificar importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about simplify.
Di un hecho verdadero sobre simplify.

Simplify ____.

Question
Pregunta

Ask a question about simplify.
Haz una pregunta sobre simplify.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about simplify.
Muestra entusiasmo sobre simplify.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with simplify.
Dile a un compañero qué hacer con simplify.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

I can add the microphones because they are the ____ type.
Puedo sumar los micrófonos porque son del ____ tipo.

I cannot mix microphones and stands because they ____.
No puedo mezclar micrófonos y soportes porque ____.

The like terms are ____ and ____.
Los términos semejantes son ____ y ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which pair are like terms?

A. 6x and 2x

B. 6x and 6y

C. 6x and 6

D. 6x and $2x^2$

Show your work:

2. Simplify 5a + 3 + 2a.

A. 7a + 3

B. 10a

C. $7a^2 + 3$

D. 52a + 3

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Three students simplified $4x + 3 + 2x + 5 + x$ differently. Student A got $7x + 8$. Student B got $7x^2 + 8$. Student C got $6x + 9$. Who is correct? Explain the mistakes the other two students made.

Sentence starter: Student ____ is correct because $4x + 2x + x =$ ____x and $3 + 5 =$ ____. Student ____'s error was ____. Student ____'s error was ____.

Show your work:

Reflect — Exit Ticket

Simplify $6x + 3 + 2x + 5$.

A. $8x + 8$

B. $8x^2 + 8$

C. $62x + 35$

D. $16x$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 6x and 2x *— 6x and 2x are like terms because they both have the same variable (x) raised to the same power (1).*
2. **Try It 2:** A. 7a + 3 *— 5a + 2a = 7a. The constant 3 stays. Result: 7a + 3.*
3. **Exit Ticket:** A. 8x + 8 *— 6x + 2x = 8x and 3 + 5 = 8, so the simplified expression is 8x + 8.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Simplify is to write an expression in its shortest form.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).